

Distilling Context into Parameters for Time-Series Foundation Models for Transportation Forecasting

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I. BACKGROUND

Forecasting transportation dynamics is central to intelligent transportation systems. Time-Series Foundation Models (TS-FMs) such as Chronos-2 have recently shown strong zero-shot performance across diverse transportation datasets, often matching domain-specific deep learning without task-specific training [1]. However, these TS-FMs rely on long context windows at inference, requiring high memory and increasing latency. As TS-FMs handle increasingly long context windows, a tradeoff emerges between context richness and computational cost.

In [2], Doc-to-LoRA (D2L) is proposed as a framework addressing an analogous problem for LLMs: a hypernetwork distills long textual contexts into lightweight LoRA adapters in a single forward pass, largely preserving performance under significantly reduced inference time and memory. We propose transferring this idea to time-series forecasting: training a hypernetwork to map historical sensor data into LoRA adapters for Chronos-2, removing the need for long context windows at inference.

II. APPROACH

We seek to adapt D2L’s Perceiver-based hypernetwork architecture to Chronos-2 for transportation time series.

A. Context Distillation

Train the hypernetwork to consume a historical context window and output a LoRA adapter in one forward pass. Training minimizes the KL divergence between Chronos-2’s predictions with full context and its predictions using only the generated LoRA and a significantly smaller context. We evaluate whether this preserves forecasting quality and probabilistic calibration while reducing latency and memory.

B. Transfer Learning

Test whether a hypernetwork trained on a subset of stations (e.g., 10–20%) generalizes to unseen stations without retraining. This would mean it learns a general mapping, enabling rapid deployment to new sensors or cities.

C. Data

We build upon the benchmark of [1] using the PeMSD7 traffic dataset, which contains historical traffic speed and flow data collected by the California Department of Transportation (Caltrans) Performance Measurement System (PeMS). If time

allows, we will consider additional datasets included in the benchmark.

III. RESEARCH QUESTIONS

We formulate the following research questions, listed in the order we intend to investigate them. It is not expected that we will have time to cover all of them within the scope of this project.

- 1) Can we preserve TS-FM performance under significantly lower inference time and memory requirements?
- 2) Does the trained hypernetwork generalize to producing effective LoRA weights for stations not seen during training?
- 3) If we replace the teacher target in the KL divergence with ground-truth labels, does performance improve beyond the baseline?
- 4) Can we encode non-time-series data into the LoRA—such as weather or neighbouring station data—to improve TS-FM performance?
- 5) Does the hypernetwork generalize to generating LoRA weights for other datasets without retraining?

IV. BUSINESS IMPACT

Compressing context into a small LoRA adapter could significantly cut inference costs, making TS-FMs practical for transit agencies and mobility operators with limited infrastructure budgets.

Furthermore, if the hypernetwork generalizes across stations, operators can deploy forecasts to new locations without retraining, which is useful for expanding bike-sharing systems, new EV charging networks, or cities adopting intelligent traffic management.

Finally, this project contributes to broader research on Transformer architectures by investigating whether the mechanisms proposed in [2] generalize effectively to non-textual foundation models. If D2L performs well across transformer-architecture this could benefit a wide range of downstream machine learning tasks deployed in the real world.

REFERENCES

- [1] J. Pulido and F. Rodrigues, “Time Series Foundation Models as Strong Baselines in Transportation Forecasting,” *arXiv:2602.24238*, 2026.
- [2] R. Charakorn, E. Cetin, S. Uesaka, and R. T. Lange, “Doc-to-LoRA: Learning to Instantly Internalize Contexts,” *arXiv:2602.15902*, 2026.